

MATH222 - Linear Algebra II

1. Instructor : Euisung Park

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2. Course Overview: This course provides an introduction to linear algebra, with a focus on eigenvalues and eigenvectors, orthogonality, and positive definite matrices. We begin by studying the definitions, properties, and applications of eigenvalues and eigenvectors. We then cover orthogonal projections, the Gram-Schmidt process, and inner product spaces. Finally, we examine positive definite matrices and their applications in various mathematical problems.

3. Required Textbook: Linear Algebra, 4th edition, by Friedberg, Insel and Spence

4. Office Hour : There are no fixed office hours for this course.

If you have any questions related to the course, please contact me by email.

5. Assessment and Grading:

(1) Homework

There will be no graded homework assignments. Recommended exercise problems will be posted after the Wednesday class, and their solutions will be posted the following Monday.

(2) Quizzes (300 Points)

There will be **four Quizzes** during the semester, worth a total of **300 points**.

For the quiz schedule, please refer to the Lecture Schedule below.

(3) Midterm and Final Exams (300 points each)

There will be one midterm exam (300 points) and one final exam (300 points).

For the exam schedule, please refer to the Lecture Schedule below.

Students are expected to take all quizzes and examinations. Failure to take any quiz or examination without prior approval will result in a grade of **F** for the course.

(4) Final Grade (Total: 1000 points)

Your final grade will be determined as follows:

Attendance: 100 points, Quizzes: 300 points

Midterm Exam: 300 points, Final Exam: 300 points

Lecture and Exam Schedule

	Tuesday	Thursday	Remarks
Week 1	(9/1) Review	(9/3) Section 5.1	
Week 2	(9/8) Section 5.1	(9/10) Section 5.2	
Week 3	(9/15) Section 5.2	(9/17) Quiz 1 / TF 1	
Week 4	(9/22) Section 5.4	(9/24) No Class	추석 연휴
Week 5	(9/29) Section 5.4	(10/1) Section 6.1	
Week 6	(10/6) Section 6.1	(10/8) Section 6.2	
Week 7	(10/13) Section 6.2	(10/15) TF 2	
Week 8	(10/20) Midterm	(10/22) Section 6.3	Midterm Period

	Tuesday	Thursday	Remarks
Week 9	(10/27) Section 6.3	(10/29) Section 6.3	
Week 10	(11/3) TF 3	(11/5) Quiz 2	
Week 11	(11/10) Section 6.4	(11/12) Section 6.4	
Week 12	(11/17) TF 4	(11/19) Quiz 3	
Week 13	(11/24) Section 6.5	(11/26) Section 6.5	
Week 14	(12/1) Section 6.6	(12/3) Section 6.6	
Week 15	(12/8) Section 6.6	(12/10) Quiz 4 / TF 5	
Week 16	(12/15) TF 6	(12/17) Final Exam	Final Exam Period